Survival Analysis of Political Donations

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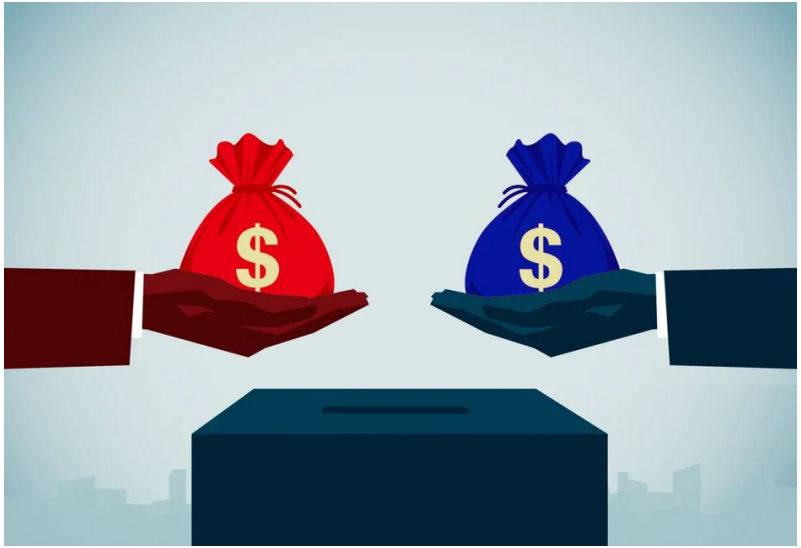
Background - What is the purpose of political donations?



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Background – Why do politicians need money?

[3] **Staffers**

[4] **Campaign Stops**

[5] **Campaign Transportation**

Federal Election Commission

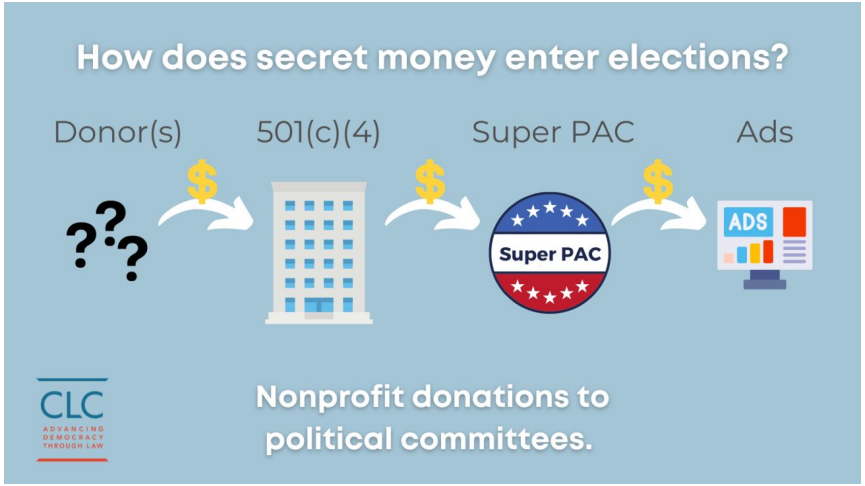
[2]

Purpose: to protect the integrity of campaign finance processes and to enforce campaign finance laws.

The FEC places limits on campaign donations.

In 2010, the FEC lost a lawsuit against Citizens United which effectively lifted the ban on corporations from spending money on independent electioneering [7].

Background - Fears around no limits on contributions



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Who is funding your campaign?

[9]

Data & Methodology

Data Collection:

Contains donation records with receipt dates, amounts, and political affiliations.

Variables:

Response Variable: Time to Election (days)

Covariates: Political Affiliation (Bipartisan, Democratic, Republican)

Experimental Units: Individual donations

Sample Size: 1510 observations

Censoring: Right-censoring present for donations not made by the election day.

- Total Donations Recorded: 1,470
- Censored Observations: 40 (Cases where full information about donation timing wasn't available)
- Censoring Rate: 2.65%

Results & Analysis

Main Questions of interest:

1. Characterize the time between the date of the donation and Nov 5, 2024 (election day)
2. Did donors tend to donate later in the election cycle or earlier?
3. How did this vary by political affiliation?

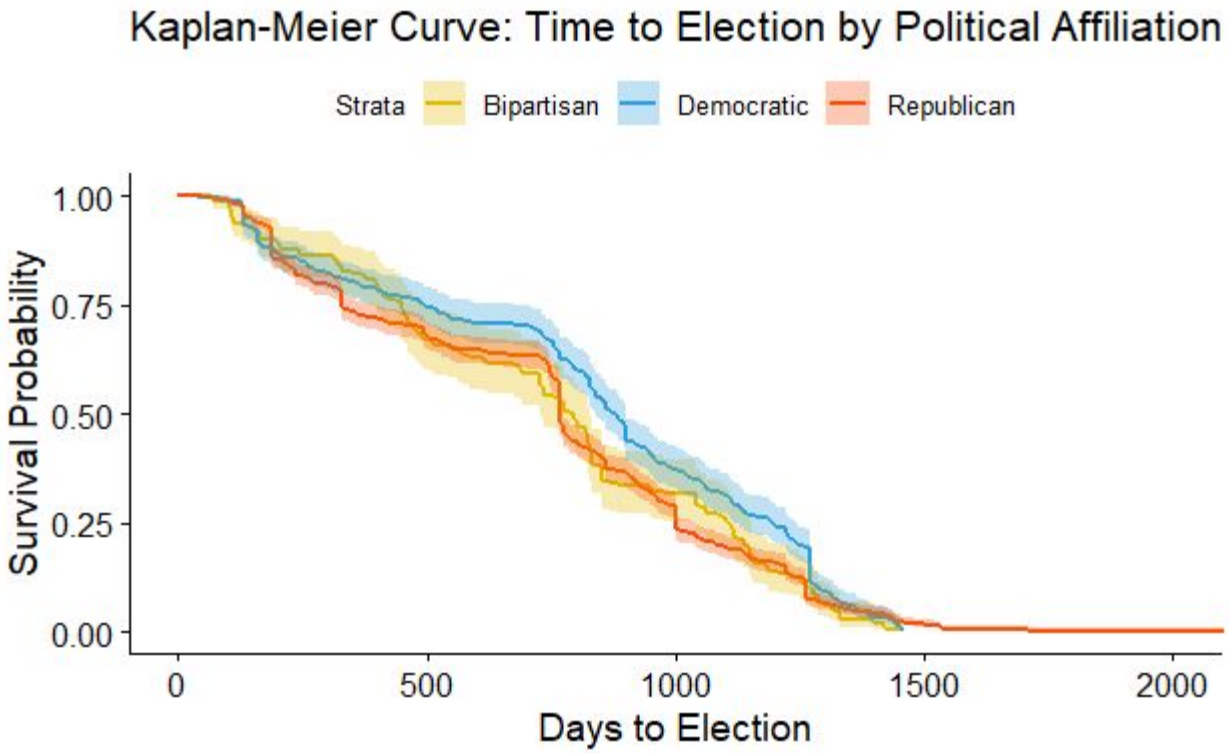
Results & Analysis

Descriptive Statistics of Political Donations

- **Donation Amounts:**
  - Minimum: $5
  - Maximum: $10,000,000
  - Mean: $83,111.74
  - Median: $3,300
  - Standard Deviation: $539,491.20 (high variability due to large donations)
- **Time to Election** (Days Between Donation and Election Day - Nov 5, 2024):
  - Minimum: 39 days
  - Maximum 1463 days
  - Mean: 765.87 days (about 2.1 years)
  - Median: 788 days
  - Standard Deviation: 403.58 days
- **Amount of donation money each party received**
  - Republican: $114,680,236 (916 donations)
  - Democratic: $5,215,693 (389 donations)
  - Bipartisan: $2,195,215 (164 donations)
  - Only 1 donation of $501 was towards an unspecified party in the dataset

Results & Analysis

- Survival probability decreases as Election Day approaches.
- Democrats and Republicans have different donation timing behaviors; Republicans tend to get donations sooner, and Democrats seem to get donations later in the election cycle.
- Bipartisan donors contribute at a different pace compared to party-affiliated donors.
- P-value from log-rank test: 0.007
- Interpretation: The differences in donation timing are statistically significant, meaning political affiliation impacts the likelihood of donating earlier or later.

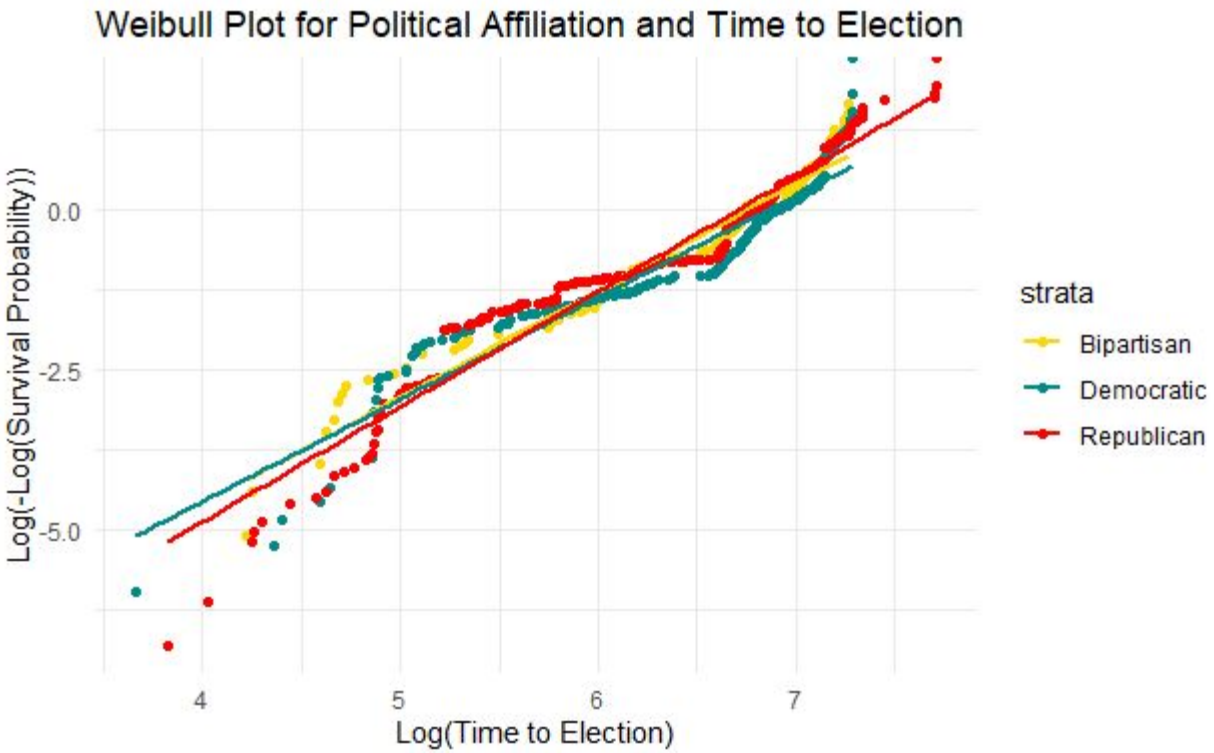


Results & Analysis

Diagnostics & Goodness-of-Fit

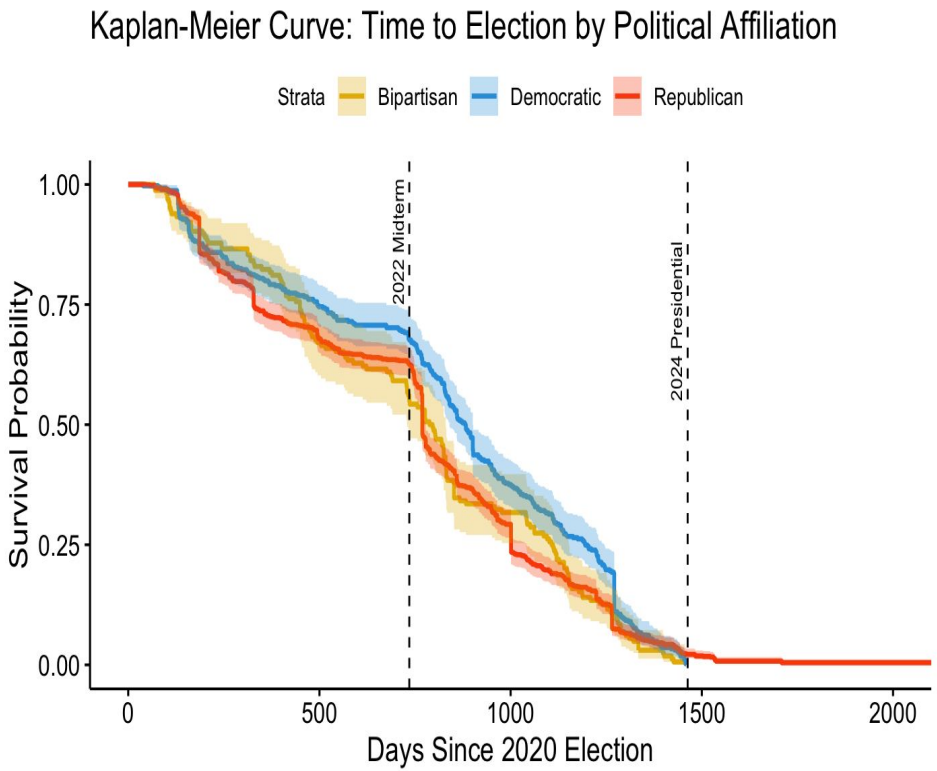
- **Weibull model chosen based on:**
- Lowest AIC (21752.19)
- Weibull plot suggests linearity in log-log survival transformation
- Reasonable parameter estimates

- **Potential Model Violations:**
- Donations may be influenced by external campaign events.
- Censoring is minimal, but data may not capture unreported small contributions.



Conclusions

- Effect of midterms on political donations
- Political actors may be trying to capitalize on any gains or mitigate any losses during midterms
- Gaining Early influence
- Differences in donation probability are probably affected by political capital held by different parties
- Understanding the timeline of when political campaigns make a push for influence



Sources

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[2] https://www.caidp.org/public-voice/fec-us-2023/

[3] https://slate.com/human-interest/2013/11/congressional-staffers-what-s-it-like-to-work-for-a-member-of-congress.html

[4] https://www.youtube.com/watch?v=a4Ejd8BcueM

[5] https://www.wesa.fm/politics-government/2020-07-20/federal-judge-agrees-to-expedite-trump-campaigns-lawsuit-over-vote-by-mail-in-pa

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[9] https://www.nytimes.com/2024/10/07/style/elon-musk-donald-trump-campaign-rally.html

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Thank You!

Questions?